Economic History II

Political identity and values

FGV EPGE

Outline of the class

- **Structural transformation and value change**: the British abolitionist movement. Figueroa and Fouka (2023).
- **Nation building through foreign intervention**: the US bombing of Vietnam. Dell and Querubín (2018).
- One question runs through both: what makes a society's values change, and what keeps them fixed.

Structural transformation and value change

Figueroa and Fouka (2023)

The British abolitionist movement

Structural Transformation and Value Change: The British Abolitionist Movement
Valentín Figueroa and Vasiliki Fouka
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JEL No. A13,N63,O14,P16,Z10

ABSTRACT

What drives change in a society's values? From Marx to modernization theory, scholars have identified a connection between structural transformation and social change. To understand how changes in a society's dominant mode of production affect its dominant values, we examine the case of the movement for the abolition of slavery in the late 18th and early 19th century Britain, one of history's most well-known campaigns for social change, which coincided temporally with the Industrial Revolution. We argue that structural transformation alters the distribution of power in society and enables groups with distinct values and weak economic interest in the status quo to mobilize for change. Using data on anti-slavery petitions, membership in abolitionist groups, MP voting behavior in Parliament and economic activity, we show that support for abolition was strongly connected to manufacturing at the aggregate and individual level. We rely on biographical data and the analysis of parliamentary speeches to show that industrialists were relatively less reliant on income from slavery and were characterized by a universalist worldview that distinguished them from established elites. Together, our findings suggest that both values and economic interest play a role in driving social change.

The puzzle

- Value systems, mappings from actions to moral valuations, vary across societies and over time.
- Dueling: in 1829 the Duke of Wellington, sitting Prime Minister, fought one. By the 1850s the practice was dead in England.
- Approval of same-sex relations among Americans went from 20% in 1990 to 60% in 2020.
- Britain, 1787 to 1833: slavery goes from a fundamental institution of the Empire to **morally repugnant**.

British slavery

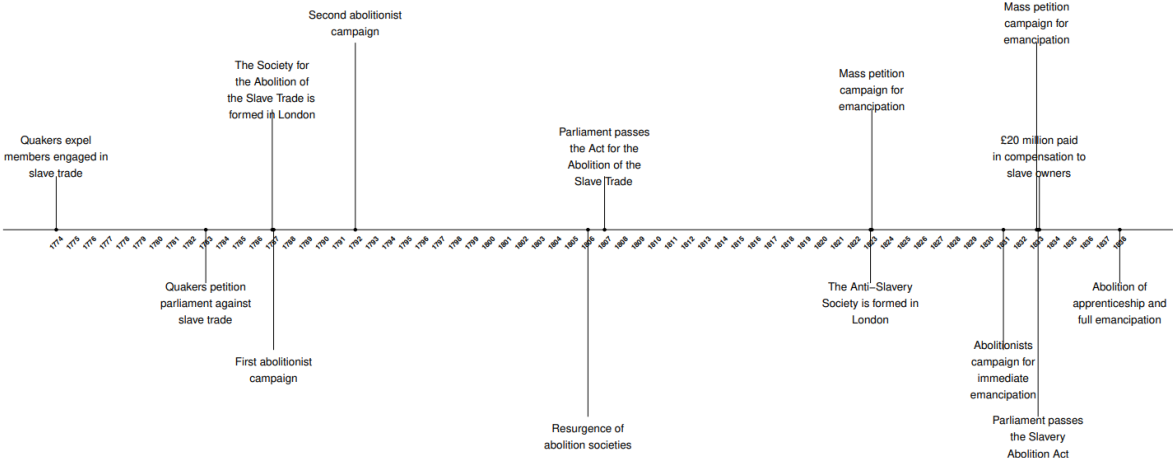
- First decade of the nineteenth century: the Atlantic slave trade carried over 60,000 enslaved persons per annum.
- Britain was the dominant player. More than half of those transported between 1791 and 1805 went on British ships.
- Slave labor was the main input for sugar. British colonies in the West Indies, Jamaica and Trinidad among them, produced more than half the world's sugar in 1805.


NORTH AMERICA
EUROPE
AFRICA
SOUTH AMERICA
ATLANTIC OCEAN
ATLANTIC OCEAN
GREAT BRITAIN 34.2%
OTHER EUROPE 0.3%
NETHERLANDS 5.8%
FRANCE 14.8%
PORTUGAL 4.2%
SPAIN 1.9%
NORTH AMERICAN MAINLAND 2.9%
CARIBBEAN 5.5%
AFRICA 0.3%
OTHER BRAZIL 0.3%
PERNAMBUCO 4.4%
BAHIA 12.7%
SOUTHEAST BRAZIL 11.5%
RÍO DE LA PLATA 0.1%
Countries and Broad Regions of the Atlantic World Where Slave Voyages Were Organized & Shares of Captives Carried Off from Africa 1501–1867
Number of slaves
2,800,000
1,200,000
500,000
100,000
25,000
Percent figures represent shares of total captives

The movement

- 1783: the **Quakers** put abolition on the parliamentary agenda, on religious grounds. Quickly dismissed.
- 1787-8: the Society for Effecting the Abolition of the African Slave Trade. Nine of the twelve founding members were Quakers.
- Method: printed material, public meetings, lecture tours, and above all bursts of petitions flowing to Parliament.
- Manchester, 1788: 11,000 signatures, a fifth of the town's population.
- 1778 to 1833: at least 12,124 petitions. In 1814, and again in 1833, 20 percent of the British adult population signed one.

Figure 1. Timeline of abolitionist movement



Abolition was costly

- The 1833 bill compensated slave owners **20 million pounds**, about 5% of British GDP.
- Kaufmann and Pape: the whole anti-slavery effort cost Britain roughly 1.8% of GDP every year for sixty years, 1808 to 1867.
- That figure includes the naval, legal and diplomatic cost of suppressing the trade, lost customs revenue, the lost sugar-carrying trade, and higher sugar prices at home.
- Also the loss of life, mainly from disease, among the Royal Navy crews enforcing the ban.

Industrialization, at the same time

- The Industrial Revolution begins in England around 1750: textiles, metallurgy, transportation.
- 1780 to 1870: real GDP per capita doubles, putting Britain 70 percent ahead of France and Germany.
- The share of the labor force in agriculture halves, to 22.7 percent. Around a third of the population works in manufacturing.
- Manchester, Birmingham and Sheffield grew more than **ten-fold** between 1750 and 1850.

Who the industrialists were

- Crouzet's sample of major industrialists, 1750 to 1850: 85% came from a middle-class background, almost half from low-middle-class families.
- In Crouzet's words, "shopkeepers, self-employed craftsmen and artisans, cultivators of various kinds".
- And: "neither the upper class nor the lower orders made a large contribution to the recruitment of industrialists".
- Often religious **dissenters**: a Protestant work ethic together with enlightened ideas about liberty. Josiah Wedgwood, Richard Reynolds.
- By the Victorian era England had two middle classes: a wealthier, capital-intensive one based on commerce in London, and a labor-intensive one based on manufacturing in the North.

The historiographical debate

- **Williams** (1944): capital from the slave trade financed early industry; then industry overshadowed sugar, the West Indies mattered less, and mercantilism became an obstacle to British exports.
- What is central in Williams is the relative, not the absolute, contribution of slavery to the British economy.
- **Davis** (1966): abolition instigated by the industrial bourgeoisie, slavery as the antithesis of the free-labor order on which their rise depended.
- **Drescher** (1986): a middle class phenomenon, artisans and miners excluded from the political system, petitioning as the extra-institutional route.

The argument

- Structural transformation, the rise of a new dominant sector, shifts the **distribution of power** in society.
- Change is initiated by ascending groups with three things at once: weak economic interest in the status quo, values distinct from entrenched elites, and enough power to act on them.
- The cost of mobilizing falls as participation rises. So once industrialists move, weaker groups with aligned values can follow.
- Both economic interest and values do work here. Neither alone gets the timing right.

$U_g = \pi\left(v_g - y_g^s\right) - c$ π: chance of success c: cost of joining

v_g: values y_g^s: slavery income y_g^o: other income $p(\cdot)$: power

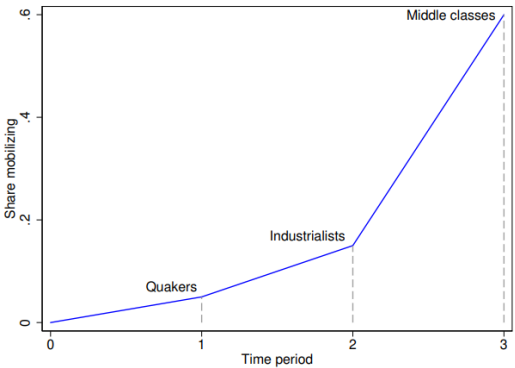
Table 1. Model parameters applied to abolitionism in Britain

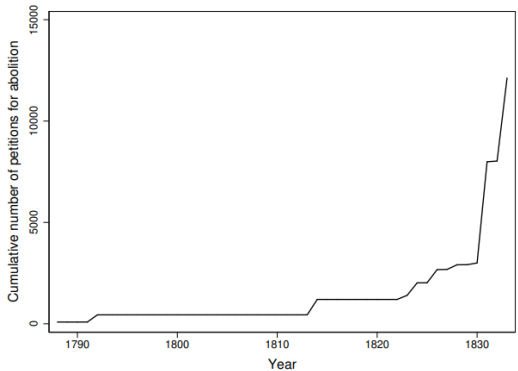
Group	v_g	y_g^s	y_g^o	$p(y_g^o + y_g^s)$
Quakers	High	Low	Low	Low
Industrialists	Intermediate	Low	Low $\rightarrow$ High	Low $\rightarrow$ High
Middle classes	Intermediate	Low	Low	Low
Landed elite	Low	Intermediate	High	High
Planters	Low	High	Low	High

The five groups

- **Quakers**: extreme anti-slavery values, low slavery income, low power. They move first on values alone.
- **Industrialists**: intermediate values, low slavery income, power moving from low to high as industry grows.
- **Middle classes**: intermediate values, low slavery income, low power. They follow once the cost of joining has fallen.
- **Landed elite**: conservative values, some slavery income, high power. **Planters**: conservative values, high slavery income. Neither joins.

Figure 2. Dynamics of mobilization against slavery



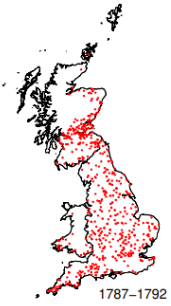


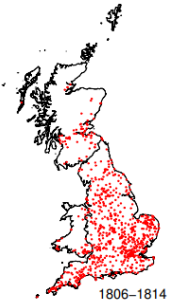
The data

- **Petitions**: topic, date and town of origin from the indices of the Journal of the House of Commons, geocoded to about 9,000 parishes in England and Wales.
- **Signatures**: the 1806 Manchester petition, one of the few documents to survive the 1834 fire at Westminster.
- **Prominent men**: everyone with a biography in the Oxford Dictionary of National Biography, matched to the 1787 Anti-Slavery Society's published subscriber list.
- **Press**: 12,421 articles mentioning slavery, 1787 to 1833, from the British Newspaper Archive.
- **MPs**: biographical dictionaries plus the Legacies of British Slavery database.

The evidence, in one slide

- **Petitions**: industrializing parishes petitioned more against slavery (coal 19.4%, mills 21%; larger with the instrument). Also for reform and Catholic rights, not against abolition.
- **Signatures**: in Manchester, manufacturers signed at 7.8%, textile workers at 0.5%. Manufacturers signed first.
- **Prominent men**: Quakers, then manufacturers, joined the 1787 Society. Manufacturers gave the most.
- **Press**: industrial districts wrote about slavery in moral terms, less as property.
- **MPs**: industrialists about 27 points more likely to vote anti-slavery in 1833, with the most universalist speeches.


1787–1792


1806–1814


1821–1829


1830–1833

Measuring industrialization

- No year-by-year local measure of industrialization exists for 1787 to 1833. The authors build two.
- **Coal**: the share of a parish over coal-bearing bedrock, multiplied by total coal consumption in England that year.
- **Water mills**: fifteenth-century mills from the Inquisitions post Mortem, multiplied by an index of national industrial output.
- Crafts and Wolf: "if there was a necessary condition for having a cotton textiles mill in 1838, a cheap source of power is the nearest candidate".

Industrialization and anti-slavery petitions

- As a parish industrialized, it sent more petitions against the slave trade and slavery.
- Coal: parishes with the largest area share over coal deposits sent **19.4%** more petitions over the period than parishes with none.
- Water mills: 21% between parishes with none and parishes with the maximum of ten.
- Both hold when parishes are allowed onto different trends by terrain, soil and location, and once Quaker meeting houses and the local gentry are in.

The worry about medieval mills

- Parishes with fifteenth-century mills may have been different before the Industrial Revolution in ways that terrain, soil and location cannot capture.
- The direction is not obvious. Earlier development could mean faster urbanization and more petitioning.
- It could also mean less turnover in the social structure, closer ties to landed elites, and **more conservative** values.
- The fix: predict where a mill could have been built from the physical environment alone.

Water power potential

$$\text{Power} = \underbrace{m}_{\text{water flow}} \times g \times \underbrace{H}_{\text{drop}} \times \eta$$

- g (gravity) and η (efficiency) are the same everywhere: potential is flow $\times$ drop.
- In the data: upstream area draining into the parish $\times$ terrain slope.
- Lots of water or steep terrain alone is not enough. It takes both.

An instrument for medieval mills

- The instrumental variable idea from Econometria I. Problem variable: medieval mills. Instrument: flow $\times$ slope, with flow and slope each held fixed on their own.
- Relevance: high flow on steep terrain did predict where mills were. Strong first stage.
- Exclusion: terrain matters for petitioning only through mills and the industry they became. Adding every parish characteristic barely moves the estimate.
- Result: an order of magnitude larger than the plain comparison. Three readings:
  - mills sat in richer, more conservative places, which hid the effect;
  - parishes with a mill only because of terrain were the ones that industrialized;
  - the medieval mill count measures water power with error.

Dams
Esther Duflo and Rohini Pande
NBER Working Paper No. 11711
October 2005
JEL No. O21, O12, H43, H23

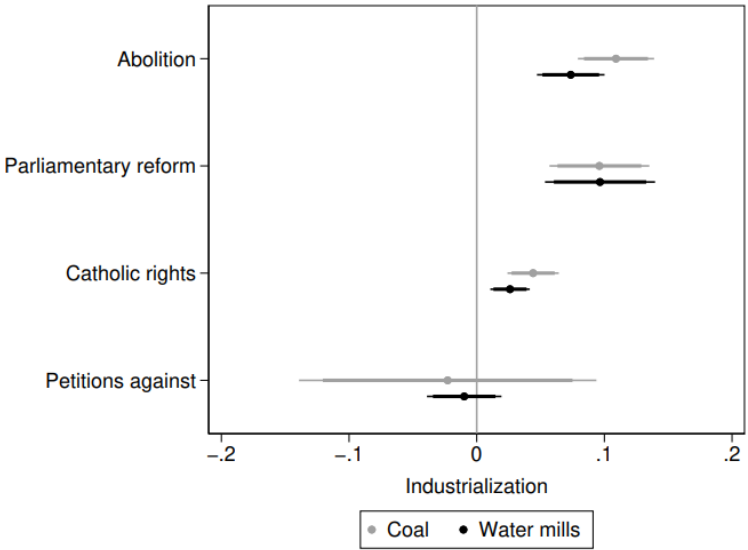
ABSTRACT

The construction of large dams is one of the most costly and controversial forms of public infrastructure investment in developing countries, but little is known about their impact. This paper studies the productivity and distributional effects of large dams in India. To account for endogenous placement of dams we use GIS data and the fact that river gradient affects a district's suitability for dams to provide instrumental variable estimates of their impact. We find that, in a district where a dam is built, agricultural production does not increase but poverty does. In contrast, districts located downstream from the dam benefit from increased irrigation and see agricultural production increase and poverty fall. Overall, our estimates suggest that large dam construction in India is a marginally cost-effective investment with significant distributional implications, and has, in aggregate, increased poverty.

The same trick: dams in India

- Problem variable: dams in a district. Governments put them where farming pays off or where local politicians have clout.
- Instrument: river gradient. Gently sloping rivers suit irrigation dams; flat rivers, like the Ganges plain, do not.
- The state decides how many dams to build. Gradient decides which of its districts get them.
- Exclusion: gradient matters for farming and poverty only through dams. Each district's own geography is held fixed.
- Result: in the dam's district, no gain in farm output and more poverty. Downstream, more irrigation, more output, less poverty.

Figure 4. Industrialization and other petitioning campaigns



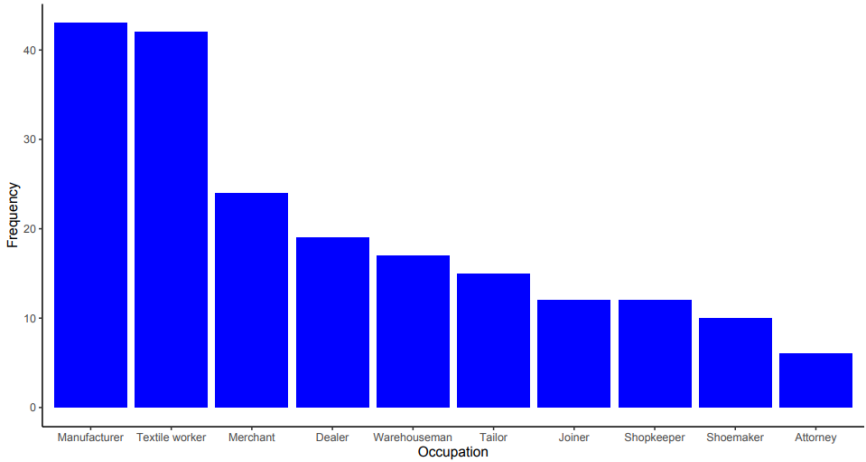
Not only abolition

- Same exercise, other petitioning campaigns of the period.
- Industrializing parishes petitioned more for **parliamentary reform**: a wider franchise, and seats moved from the rotten boroughs to towns that had grown with industry.
- And more for Catholic rights: repeal of the restrictions on Catholics and Dissenters, including on holding public office.
- Petitions *against* abolition show no positive relation to industrialization. So this is not mechanically population growth.

Who was actually petitioning?

- The parish-level pattern could be driven by the middling strata, or by the working class. Aggregates cannot tell them apart.
- The 1806 Manchester petition to the House of Lords: **2,348 signatures**, encouraged by Thomas Clarkson in reply to an anti-abolitionist petition.
- The Manchester and Lancashire Family History Society matched signatories to three trade directories: 312 unambiguous matches, 281 with an occupation listed.
- De Tocqueville on Manchester in 1835: “Here humanity attains its most complete development and its most brutish; here civilisation works its miracles, and civilised man is turned back almost into a savage”.

Figure 5. Top 10 most frequent occupations of Manchester petition signatories



The occupations

- **Manufacturers** are the most common occupation among the matched signatories: 43.
- Then textile workers, weavers, spinners and dyers: 42. Merchants 23, dealers 19, warehousemen 17, tailors 15.
- Manufacturers and textile workers signed in equal numbers, though textile workers were by far the larger group in the town.
- As a share of Manchester residents in each occupation in the 1813 census, industrialists signed at 7.8%, textile workers at 0.5%.

Signature order

- Signatures in Britain were not collected door to door. Petitions were posted in inns, shops and churches, and you walked there to sign.
- So the order in which someone signed says something about how central they were to the movement.
- The document is nine sheets of parchment stitched into a five-meter roll. Four proxies for order: page, column within page, entry within column, entry within page.
- Caveat the authors raise: notables were sometimes placed high on a list to induce others to sign. That is still consistent with the argument.

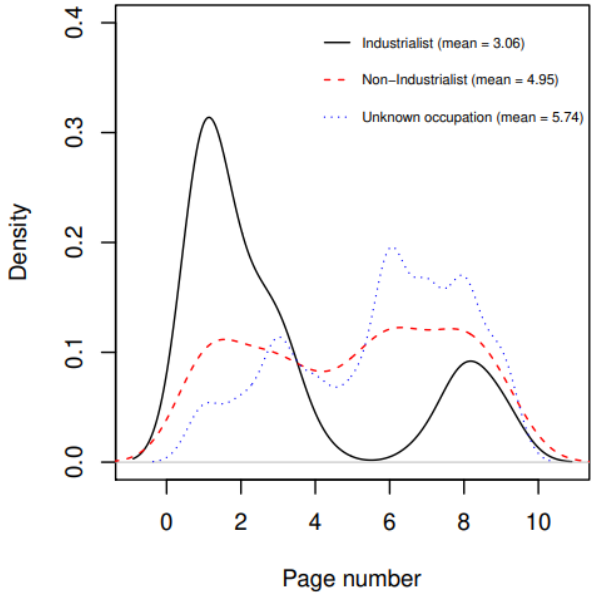
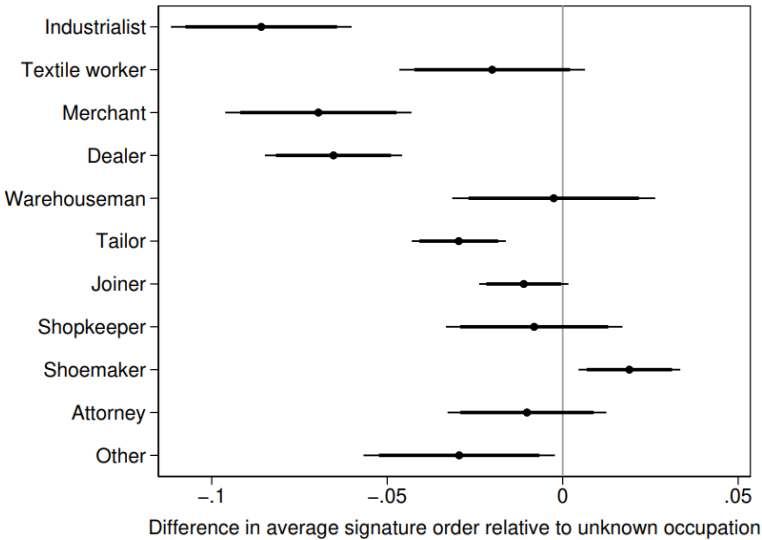

Industrialist (mean = 3.06)
Non−Industrialist (mean = 4.95)
Unknown occupation (mean = 5.74)
Density
0.0
0.1
0.2
0.3
0.4
0
2
4
6
8
10
Page number

Figure 7. Effect of occupation on signature order



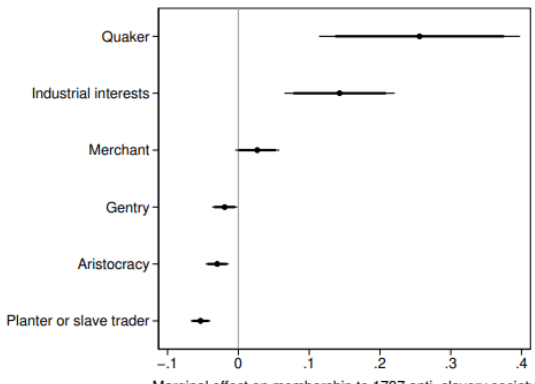
Manufacturers signed early

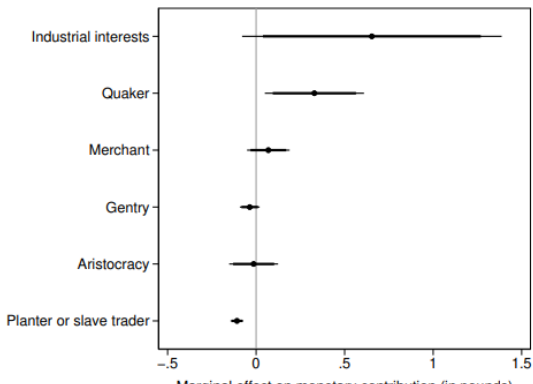
- Across all four measures the distribution for manufacturers sits to the left of the others.
- Page number: an average of 3.06 for industrialists, 4.95 for other known occupations, 5.74 for signatories whose occupation is unknown.
- By occupation, industrialists and merchants are the two categories most likely to sign first. Textile workers come later.
- Middle class elites head this campaign; the industrial workforce does not.

The Anti-Slavery Society of 1787

- 3,463 men have biographies in the Oxford Dictionary of National Biography and had reached adulthood by 1787.
- The Society published its full subscriber list, with each member's monetary contribution. **162** of the dictionary's men appear on it.
- Biographies coded for religion, social rank and economic interest: Quaker, aristocrat, gentry, industrialist, merchant, planter or slave trader.

Figure 8. Membership and contributions to the Anti-Slavery Society in 1787





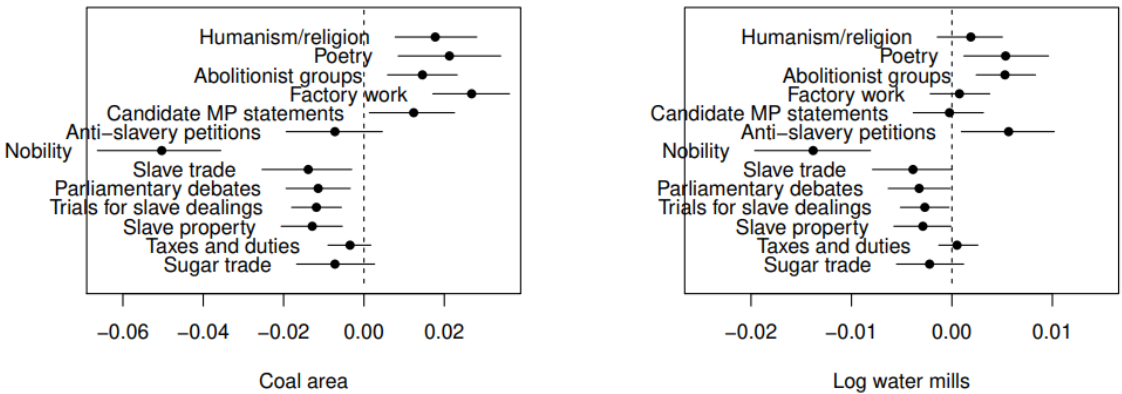
Who joined, and who paid

- Being a **Quaker** is the strongest predictor of membership among prominent men.
- Involvement in manufacturing comes next, and is the only other characteristic that predicts membership.
- On amounts contributed, manufacturers give the most, consistent with the highest income among groups holding anti-slavery values.
- Planters and slave traders join less and give less. The gentry and the aristocracy join less.

Values or interest? Read the press

- 12,421 articles mentioning slavery, published in 121 newspapers, placed at the level of the census registration district (625 districts).
- The authors sort the articles into 30 recurring themes and compare how prevalent each theme is in industrial and non-industrial districts.
- The question is whether slavery was discussed there in **moral** terms or in economic ones.

Figure 9. Topics of articles on slavery by location of publication



What industrial districts wrote about

- More prevalent in industrial districts: abolitionist groups, anti-slavery petitions, and the two topics the authors label Humanism/religion and Poetry.
- Less prevalent: slave property, the legal and financial side of owning people, and neutral references to the slave trade as commerce.
- Taxes and duties on sugar appear at similar rates everywhere.
- Not a matter of editorial line: newspapers in more industrial districts were not more liberal.

Humanism and religion

- April 1833, a letter from the Society of Friends reprinted in several papers: “Thou shalt love thy neighbor as thyself, under which term we believe are comprehended our fellow creatures of every nation, tongue and colour.”

- November 1832, a candidate MP’s statement: “The right of the slave to be raised from the level of the brute creation, to which he has been sunk through no fault of his own, and to be restored to that of human nature, in which he was created by his Maker”.

The topic we label "Poetry", is dominated by poems against slavery, frequently imbued by the same humanist principles and condemning the institution as brutal and repugnant. An example from the Derby Mercury published in April 1823, is titled "Appeal to Britain – On the Slave Trade" and reads as follows:

> Though thy name may be quoted again and again,
> As a nation of freemen, a kingdom of men;
>
> Yet a stain will be on thee, will tarnish thy name,
> And a spot of dishonour will sully thy fame,
> The blood of the slave thy fair honour shall stain.

This particular poem appealed to British exceptionalism as well as to a sense of honor, which scholars like Drescher (2002) and Appiah (2011) have identified as crucial for the change in attitudes on slavery.[32]

Poetry

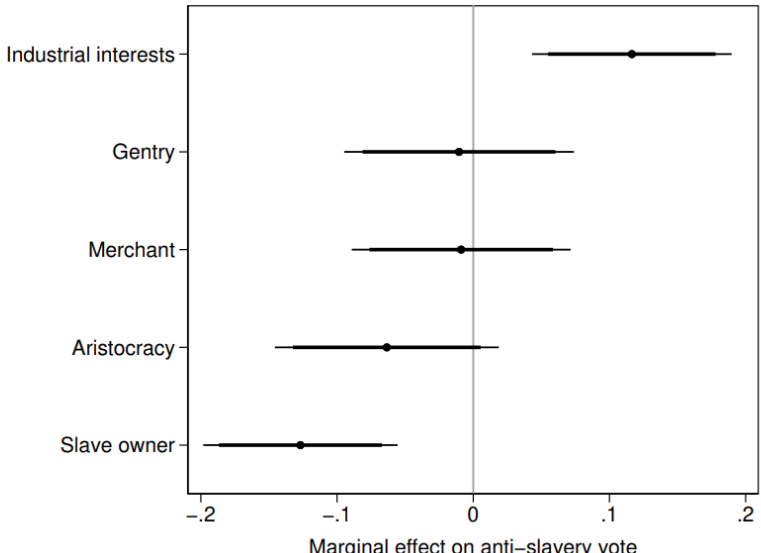
- The Derby Mercury, April 1823, under the title “Appeal to Britain”, on the slave trade.
- The appeal is to British exceptionalism and to a sense of **honor**, which Drescher and Appiah identify as crucial to the change in attitudes on slavery.
- Abolition was cast as a way for the Empire to display moral virtue after the loss of the American colonies.

Inside Parliament

The 1833 vote

- The 1832 reformed Parliament debated the Ministerial Plan for the Abolition of Slavery, enacted as the Slavery Abolition Act of 1833.
- An MP counts as anti-slavery if he voted at least once for limiting the apprenticeship period and for minimal compensation to planters.
- MPs with industrial interests were **36.2 percentage points** more likely to vote that way. About 27 points with the full set of individual and constituency characteristics.
- Slave owners voted against. Liberal and Radical leanings predicted support, as did Protestant dissent. Oxbridge, merchant standing, gentry and aristocracy did not.

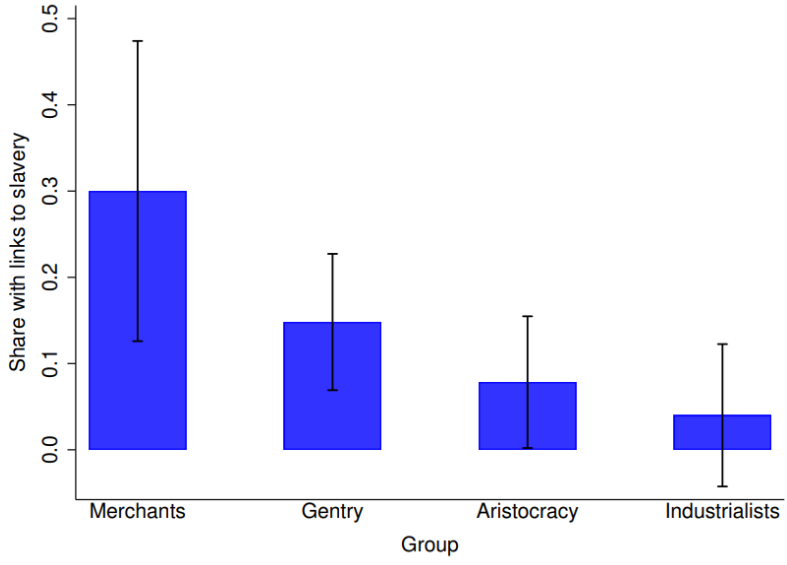
Figure 10. Support for abolition by group of MPs



The ranking of groups

- Industrialists the main supporters of abolition, slave owners its main opponents.
- The nobility less supportive, though the estimate is not quite sharp enough to call. Merchants and the gentry show no differential position.
- Earlier debates point the same way. In 1806-07, three of the four industrialist MPs who spoke supported abolition, against 30 of 46 other MPs.
- The one industrialist who opposed in 1806 was Sir Robert Peel, whose fortune came from cotton-spinning.

Figure 11. Links to slave income among British MPs



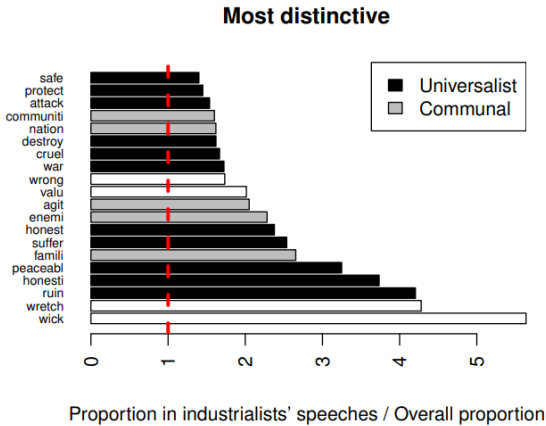
Direct links to slavery

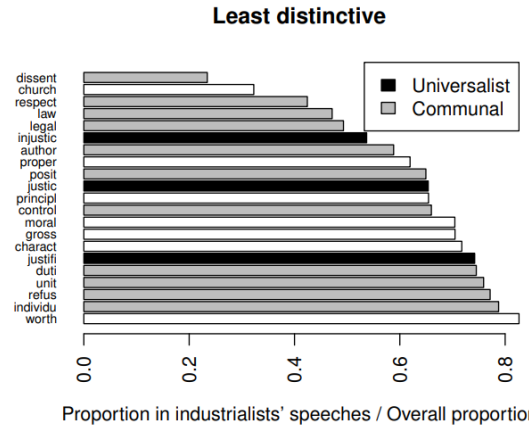
- 51 of the 550 elected MPs appear in the Encyclopaedia of British Slave-ownership, as owners or otherwise linked to West Indian estates.
- Industrialists have the weakest links: **one** manufacturer in the entire list.
- Merchants 30%, the gentry 15%, MPs with a nobility title 7%.
- If slavery income were the whole story, merchants and gentry should oppose abolition more than the nobility. The votes show the opposite.

Measuring values from speeches

- Moral Foundations Theory partitions moral concerns into foundations, each with a different principle for what is right.
- **Universalist**: non-harm and care, fairness and reciprocity. **Communal**: ingroup loyalty, respect for authority and tradition.
- Universalism is not more altruism. It is altruism that declines less rapidly with social distance.
- Consistent with that: abolitionist manufacturers tolerated exploitation in their own factories, and abolitionists placed freed slaves near the bottom of society, a "grateful peasantry".

Figure 12. Words distinctive of industrial MPs

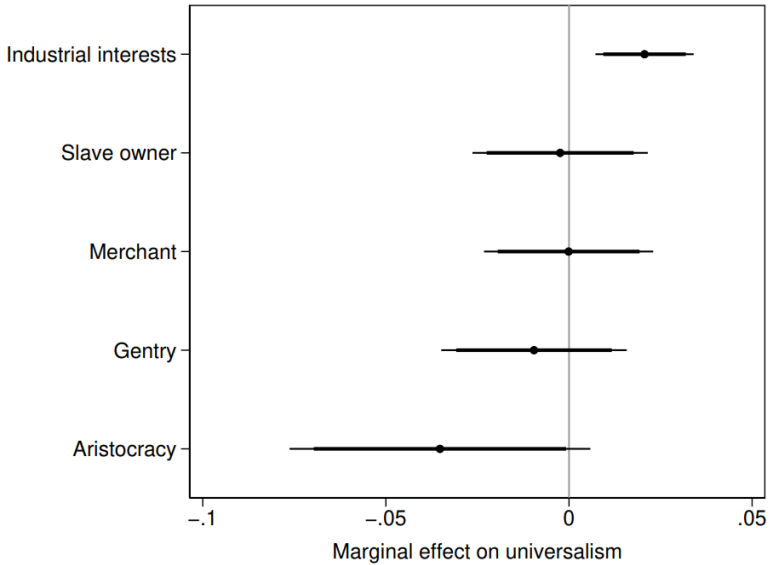




Words distinctive of industrialist MPs

- Distinctiveness: how much more often a word appears in industrialist speeches than in parliamentary speech on average.
- Most distinctive: safe, protect, cruel, wrong, suffer, honest, peaceable. The words characteristic of industrialists lean **universalist**.
- Least distinctive: church, respect, law, authority, duty, control. The words least frequent in their speeches, relative to overall use, reflect communal foundations.
- The corpus is 7,798 speeches from 1833-34, of which 368 were delivered by industrialist MPs.

Figure 13. Universalism by group of MPs



Universalism by group

- Industrialists clearly more universalist. The nobility clearly more communal. Gentry, merchants and slave owners in between.
- Not party, religion or education: neither Liberal MPs nor Protestant dissenters score higher.
- Not a topic effect either. It holds when slavery debates are dropped, and when each of 1,002 separate debates is accounted for on its own.
- Slave owners are no less universalist than the rest, which cuts against values simply adjusting to interest.

Alternatives the authors rule out

- **Direct material gain**. Adam Smith called slavery inefficient: "A person who can acquire no property, can have no other interest but to eat as much, and to labor as little as possible". But positions on free trade and the Corn Laws do not predict abolition votes.

- **Raw materials**. Textile manufacturers, dependent on slave-grown cotton, vote no differently from other industrialists.

- **Constituency pressure**. Neither petitioning in the constituency nor running unopposed changes anything. At least 130 elected MPs pledged to abolish in the anti-slavery journal *The Tourist*, and industrial interests do not predict who pledged.

- **Deflecting grievance**. Labor unrest and the Swing riots relate to anti-slavery votes in general, but no more so for industrialists.

Conclusions

- Support for abolition was strongly connected to manufacturing, at the aggregate and at the individual level.
- Opposition came from two directions: high reliance on slavery income among planters, conservative values among the old aristocracy.
- The framework accommodates Williams, Davis and Drescher rather than choosing among them.
- The open question the authors leave: **why** industrialists were more universalist than everyone else.

The same question elsewhere

- Masera and Rosenberg: the loss of comparative advantage in cotton drove the decline of pro-slavery attitudes in the US antebellum South.
- Seyler and Silve: electoral support for emancipation in **Brazil** depended on how far a district relied on slavery against immigration as an alternative source of labor.
- Brazil abolished in 1888, last in the hemisphere, and paid no compensation to owners. Britain paid 5% of GDP.

Nation building through foreign intervention

Dell and Querubín (2018)

The US bombing of Vietnam

NATION BUILDING THROUGH FOREIGN INTERVENTION: EVIDENCE FROM DISCONTINUITIES IN MILITARY STRATEGIES*

MELISSA DELL AND PABLO QUERUBIN

This study uses discontinuities in U.S. strategies employed during the Vietnam War to estimate their causal impacts. It identifies the effects of bombing by exploiting rounding thresholds in an algorithm used to target air strikes. Bombing increased the military and political activities of the communist insurgency, weakened local governance, and reduced noncommunist civic engagement. The study also exploits a spatial discontinuity across neighboring military regions that pursued different counterinsurgency strategies. A strategy emphasizing overwhelming firepower plausibly increased insurgent attacks and worsened attitudes toward the U.S. and South Vietnamese government, relative to a more hearts-and-minds-oriented approach. *JEL Codes*: F35, F51, F52

> Never before did the people of Vietnam, from top to bottom, unite as they did during the years that the U.S. was bombing us. Never before had Chairman Ho Chi Minh's appeal—that there is nothing more precious than freedom and independence—gone straight to the hearts and minds of the Vietnamese people.
>
> —Tran Quang Co

> The solution in Vietnam is more bombs, more shells, more napalm.
>
> —General William DePuy

The war

- Geneva Accords, 1954: Vietnam divided at the 17th Parallel until nationwide elections in 1956. The elections were never held, in part because the US concluded Ho Chi Minh would win by a landslide.
- A Soviet-backed communist state in the north, the US-backed Ngo Dinh Diem in the south, and a communist insurgency in the south led by the **Viet Cong**.
- 1965: Johnson deploys around 200,000 troops. Peak of over half a million in 1969. Withdrawal in January 1973.
- The Vietnamese government estimates more than 3 million deaths between 1954 and 1975, over 2 million of them civilians. A 2008 study puts the toll at 3.8 million.

Two strategies

- **Overwhelming firepower**: air strikes, artillery, search-and-destroy raids. Reduce insurgent forces, disrupt operations, crush morale.
- **Hearts and minds**: security by embedding soldiers in communities, development aid, small units, limited firepower.
- Both were considered. In practice firepower dominated. Westmoreland described his strategy in one word: firepower.
- General DePuy: "The solution in Vietnam is more bombs, more shells, more napalm".
- The case for it, deterrence. Bundy to Johnson: reprisals "will set a higher price for the future upon all adventures of guerrilla warfare".

The skeptics

- Insurgents are hard to locate. If the ordnance lands on civilians instead, it creates grievances that recruit for the other side.
- It can also widen the **legitimacy gap** between the insurgents and the South Vietnamese government.
- James Scott: a coercion-oriented approach is ill-suited to gaining cooperation, because citizens have many ways to undermine a state they do not genuinely support.
- Tran Quang Co: "Never before did the people of Vietnam, from top to bottom, unite as they did during the years that the U.S. was bombing us".

Kalyvas: why indiscriminate violence backfires

- Kalyvas, *The Logic of Violence in Civil War* (2006): violence deters defection only when it is **selective**, aimed at those who actually help the enemy.
- Selective violence leaves civilians a way out: comply and you are safe. Indiscriminate violence removes it, and siding with the insurgents can become the safer bet.
- Selecting needs information, and information comes from locals willing to denounce. They do so where one side already has control.
- Bombing from the air is the extreme case: no informers, and no way to tell an insurgent from a civilian.

The scale

- More than twice as many tons of explosives as in the Second World War. Four times more on South Vietnam than on the North.
- About **500 pounds of ordnance per inhabitant** of South Vietnam, equal in power to 640 Hiroshima-sized atomic bombs.
- 3.4 million combat sorties in Southeast Asia between 1965 and 1972, around 125,000 of them by B-52s. Around 400,000 tons of napalm.
- The guava cluster bomb carried 650 bomblets of 300 steel pellets each, 200,000 fragments per bomb.
- In the study's sample, 21% of hamlet-months had ordnance dropped within 5 kilometers of the hamlet centroid.

The rules of engagement

- Directly targeting civilians violated the laws of war. The rules still allowed many situations in which civilians could be hit.
- If any fire came from the vicinity of a village, it could be attacked without warning. **Free-fire zones**: anyone could be fired on.
- Warnings came as blanket leaflet drops over a district, too general for villagers to tell whether they were on a target list.
- The common F-105 bomber had a circular error probability of 447 feet: half the bombs fell within that radius of the target. Enough to hit a village, not enough to hit an insurgent rather than a civilian.



Đồng bào có thể bảo vệ được mạng sống của đồng bào và gia đình bằng những hành động sau đây :

+ Yêu cầu Việt Cộng ra khỏi xóm làng.

+ Yêu cầu Việt Cộng đừng núp trong thôn xóm và bắn súng vì đó là lý do của các cuộc oanh tạc.

+ Yêu cầu Việt Cộng đừng tập họp đồng bào tham dự mit-tinh vì đồng bào có thể bị chết chóc vì bom đạn.

Leaflet No. HQ-18-67 text:

To The People Who are in the Areas Temporarily Occupied by the Viet Cong

It is regrettable that the Government of Vietnam has to use bombs and artillery to drive the Viet Cong from places where they're hiding. In order to liberate your area, sometimes there is no other means. To protect your lives the government asks you to follow these measures:

- Do not live close to where the Viet Cong are gathering.
- Do not attend meetings hosted by the Viet Cong.
- Do not work for the Viet Cong.

Dear citizens. You can protect you and your family by taking the following actions:

- Ask the Viet Cong to leave the village.
- Ask the Viet Cong to stop hiding in the village and firing weapons; that is the reason for Government bombing.
- Ask the Viet Cong not to gather people for meetings because they may get killed by bombs and artillery.

Pulitzer Prize–winning novel: “We are all guilty until proven innocent, as even the Americans have shown. Why else do they believe everyone is really Viet Cong? Why else do they shoot first and ask questions later?” (Nguyen 2015, loc. 2902). Turse (2013, p. 55) argues:

> At every turn, the onus was put on Vietnamese civilians to actively demonstrate that they were indeed noncombatants... by staying out of off-limits areas (the borders of which they might not know)... by not running or not walking in a certain way, or not standing still and thus looking unnatural; by somehow forcing armed guerrillas from their villages but also not carrying weapons, which would automatically brand a Vietnamese as VC.

The question, and why it is hard

- Did bombing near population centers weaken the insurgency or strengthen it? And did it build a state that could stand on its own?
- Military forces target places where insurgency is already on the rise, so a raw comparison of bombed and unbombed places is uninformative.
- An unconditional lottery would be unfeasible, unethical, and also uninformative: resources go where they are believed to work.
- What is needed is variation **at the margin**, among places that had all been deemed potential targets.

McNamara and the whiz kids

- McNamara: economics degree at Berkeley, Harvard MBA, statistician for the Army Air Forces in the Second World War, then Ford. Secretary of Defense from 1961.
- He brought in the "whiz kids", largely from RAND and largely **economists**: Charles Hitch, head of RAND's economics division, as Comptroller; Alain Enthoven, MIT economics PhD, running Systems Analysis.
- Hitch and McKean, *The Economics of Defense in the Nuclear Age* (1960): defense as a problem of allocating scarce resources.
- Field data were key-punched into mainframes in Saigon and Washington and used to allocate resources. The tapes survive only because they were **subpoenaed** in a lawsuit against IBM.

Input: security

From 1969, US advisers and South Vietnamese officials answered **169** questions on every hamlet, every month or quarter. Some of them:

- Do enemy forces physically control this hamlet?
- Were armed enemy military forces present in inhabited areas of this hamlet during the month?
- Did the enemy initiate action against local security forces in or near the hamlet during the month?
- Were any attempts at selective terrorism (kidnapping or assassination) directed at ... local leaders of the hamlet during the month?
- Were any enemy propaganda meetings held or was printed propaganda distributed in this hamlet during the month?
- Does the enemy collect taxes from hamlet households?
- Were enemy military activities the direct cause of civilian casualties among hamlet residents?
- Were friendly military activities the direct cause of civilian casualties among hamlet residents?
- Do local residents that are prime VC targets ... sleep in their homes at night?
- How active is the PSDF (the hamlet's self-defense militia) in this hamlet?

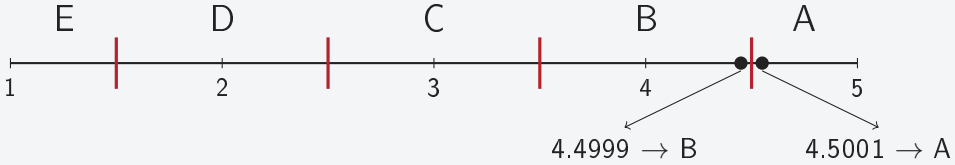
Input: government and daily life

- Did any US advisory personnel visit this hamlet during the month?
- Is there a GVN hamlet chief for this hamlet? *No / yes, appointed / yes, elected.*
- How often does the GVN village chief visit this hamlet?
- During hours of darkness, do GVN authorities enforce the laws in this hamlet?
- Have there been incidents of personal misconduct involving local security forces and hamlet residents during the month?
- Is there a market in this hamlet?
- Do any households in this hamlet own motorized vehicles (such as motorcycles, motorized sampans, cars, etc.)?
- Do any households in this hamlet own a TV set?
- Are GVN sanctioned medical services accessible to hamlet residents?
- Are the children of hamlet residents able to attend primary school classes?

Answers on fixed scales, such as *no / once / sporadically / repeatedly*.

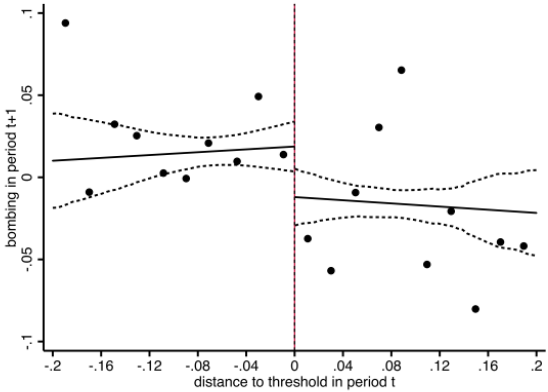
Output: one grade per hamlet, **A** (very secure) to **E**, sent to military planners.

The threshold

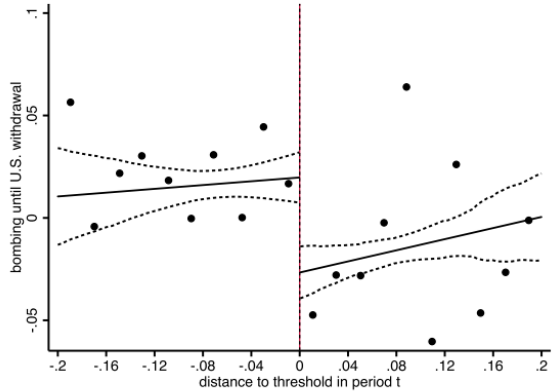


- In between, a convoluted machine: Bayes's rule with probabilities guessed by field officers, 19 sub-scores from 1 to 5, each **rounded** to a letter, then combined two or three at a time up a tree. It took the world's most powerful computer. Planners saw only the final letter.
- Two almost identical hamlets could get different letters. The one rounded down looked less secure, and was more likely to be bombed.
- The authors rebuilt the hidden scores from the archived answers and compare hamlets just below a red line with hamlets just above it.

(A) Immediate First Stage

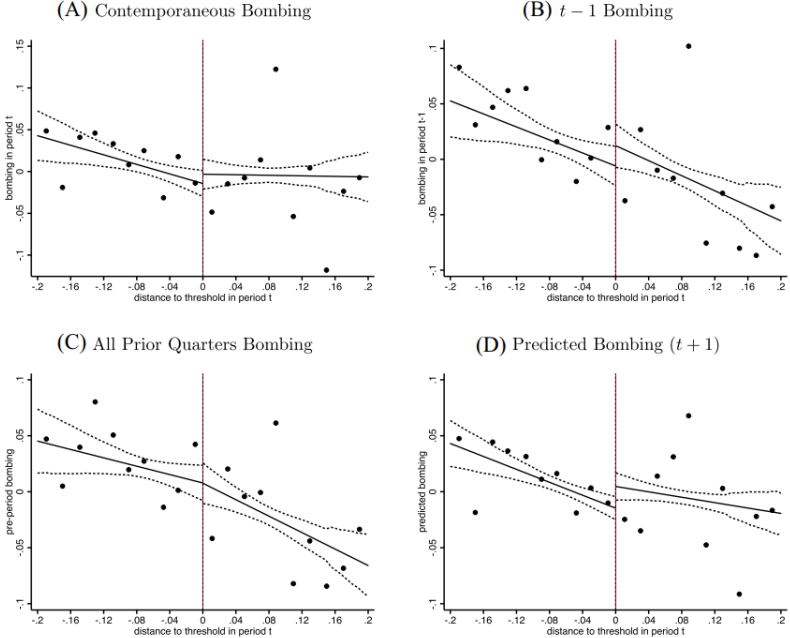


(B) Cumulative First Stage



Rounded down, then bombed

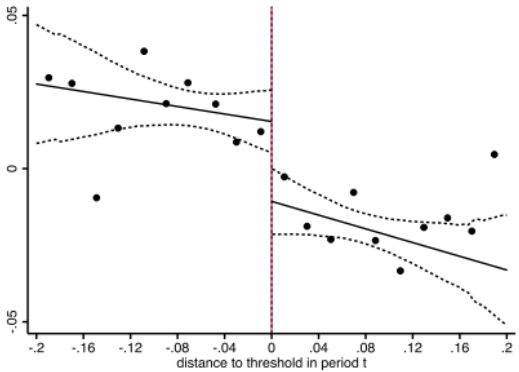
- Horizontal axis: distance to the nearest rounding threshold. Negative means the hamlet was rounded down.
- Vertical axis: strikes near inhabited areas, in the following quarter on the left, cumulated to US withdrawal on the right.
- Strikes drop at the threshold. Being rounded down made a hamlet more likely to be hit.
- Around the A-B threshold, low As had about a 5% chance of being hit in a given month, high Bs about 10%. Around D-E, low Ds about 55%, high Es nearly 75%.


(A) Contemporaneous Bombing
bombing in period t
distance to threshold in period t
(B) $t-1$ Bombing
bombing in period t-1
distance to threshold in period t
(C) All Prior Quarters Bombing
pre-period bombing
distance to threshold in period t
(D) Predicted Bombing $(t+1)$
predicted bombing
distance to threshold in period t

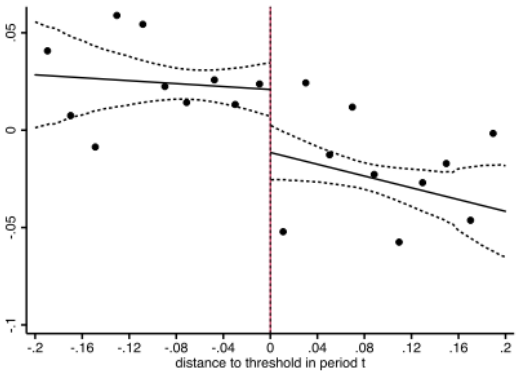
The placebos

- The data feeding the score arrived only at the close of the quarter, so there should be no contemporaneous effect. Panel A: smooth.
- Strikes in the previous quarter, and averaged over the whole period before: also smooth.
- Strikes predicted from the variables that enter the score but not from the score itself: smooth.
- In 1969 the score was computed but not disseminated. In that pilot year, being below the threshold did nothing.

(A) *VC Presence (Cumulative)*



(B) *Active VC Infrastructure (Cumulative)*



Bombing raised insurgent activity

- Moving from no strikes to the sample average raised the probability of an armed VC presence by 15 percentage points, from a mean of 0.19.
- The probability of an active **village guerrilla squad** rose by 27 points, from a mean of 0.38. That squad consists of locals, so this is not a reallocation of insurgent troops.
- The VC political branch, the VC infrastructure, was active 25 points more often.
- VC-initiated attacks on local security forces, officials and civilians rose by 9 points, from a mean of 16 percent of hamlet-months.
- VC extortion of residents rose by 23 points, from a mean of 0.27.

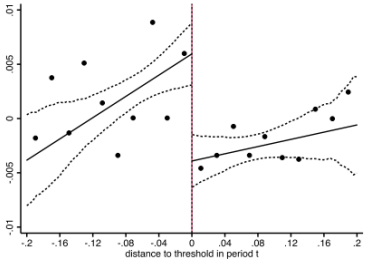
Why it could go this way

- Rank-and-file VC were part-time fighters operating near their own villages, indistinguishable from civilians without close local cooperation.
- Grievance. One soldier, after an American bomb killed his fiancee in Haiphong: he decided he had to go south and fight.
- Economic destruction lowers the opportunity cost of joining the insurgency.
- RAND interviewed 2,400 VC defectors and prisoners. Volunteers were more likely than forced draftees to report grievances against the government and economic hardship.

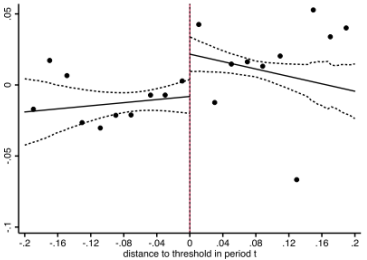
The loop

- The initial deterioration in security entered the next quarter's score, lowering it and raising the chance of being bombed again.
- That is why the cumulative effect is strong: bombing worsens security, which lowers the score, which brings more bombing.
- Ground troops followed. Friendly forces operated nearby 17 points more often over the course of the war, and US-initiated attacks rose.
- Spillovers to neighboring hamlets are limited, and where they exist they run in the same direction as the direct effects.

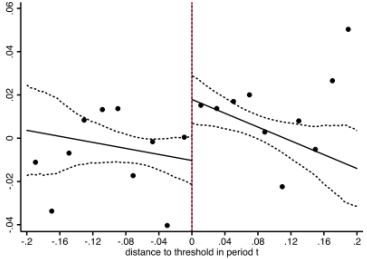
(C) *U.S. Initiated Attacks (Cumulative)*



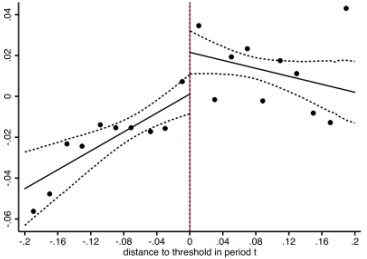
(D) *Village Committee Filled (Cumulative)*



(E) *Access to a Primary School (Cumulative)*



(F) *Participation in Civic Orgs (Cumulative)*



What bombing did to the state

- The probability that the local government systematically collected taxes fell 25 points, from a mean of 0.70.
- The village committee, which administered public goods, had all its positions filled 21 points less often, from a mean of 0.84.
- Access to primary school fell 16 points. Secondary school fell too.
- Household participation in civic organizations fell 13 points, from a mean of 0.29.
- No forced urbanization either: population growth in bombed hamlets is not measurably different.
- Scott's point, in the data: citizens have many ways to undermine a state they do not support, well short of joining an insurgency.

Conclusions

- Bombing increased VC military and political activity, weakened local government administration, and lowered noncommunist civic engagement.
- The mechanisms the authors point to are grievances and a widening of the political legitimacy gap.
- Kennedy, in 1954: “I am frankly of the belief that no amount of American military assistance in Indochina can conquer an enemy which is everywhere and at the same time nowhere”.
- McNamara, in private: “this [expletive] bombing campaign, it’s been worth nothing, it’s done nothing”.

In this class

- **Figueroa and Fouka** (2023): industrialization shifted which groups could act on their values, and slavery became repugnant.
- **Dell and Querubín** (2018): firepower applied near civilians raised insurgent activity and weakened the state it was meant to build.
- Common thread: material conditions, the distribution of power, and the values that follow from them.